

Satellite Image Processing for Urban Expansion Monitoring

Problem Statement

Urban expansion is quickly turning natural land, such as farmland, forests, and open spaces, into developed areas. This growth puts more pressure on the environment and resources. It causes problems like losing green spaces, shrinking water bodies, and increased pollution. Traditional monitoring relies on manual comparison of images from satellites. This is slow, labour-intensive and error-prone. It does not cope well with large sets of data and is not capable of delineating temporal change. We need an AI-based automated system for processing satellite imaging and automatically identifying urban expansion using image processing techniques. Using models like YOLOv8, the system now generates faster and more uniform results than previously.

Motivation

- Cities tend to rise at a fast pace and normally at the cost of nature and greenery, and thus monitoring these changes is of paramount significance to ensure sustainable progress.
- The comparison of satellite pictures by hand is slow and error-prone. AI image processing makes it faster, automatic, and precise.
- A small project like this gives us a chance to use AI and image processing in practice on an everyday problem that affects urban planning and the environment directly.

Objectives

- To preprocess and prepare satellite images for detecting urban features.
- To apply YOLOv8 object detection for identifying buildings and constructions from images.
- To analyze changes in urban areas by comparing images from different time periods.
- To generate accurate results that help in understanding the pattern of urban expansion.

Background

The cities have been growing rapidly due to increases in population, industrial developments, and expansion of infrastructure. The increase mostly yields environmental problems such as deterioration of green spaces, degradation of water bodies, and rises in the level of pollutants.

Satellite imagery provides a useful way of following such changes over a period because it consists of high-level observations of land use and urban developments. The problem, however, lies in interpreting such imagery manually because of the amount of information and complexity of urban landscapes.

Future advances in Artificial Intelligence, specifically object detection models like YOLOv8, have now made it possible to automatically detect building and construction from satellites. By comparison of photographs from different years, we get to trace how cities have been expanding and how they have been impacting the environment.

This mini-project is regarding the usage of satellite imaging and YOLOv8 image processing for urban expansion detection and generating accurate results for supporting city growth patterns

Related Work

No.	Paper Title	Method / Technology Used	Study Area / Dataset	Main Findings
1	Detecting Urban Changes with Recurrent Neural Networks from Multitemporal Sentinel-2 Data	RNN + U-Net	Sentinel-2 satellite images (Europe)	Achieved >95% accuracy in detecting urban growth using temporal patterns in data.
2	Satellite Image Classification with Deep Learning	CNN (Convolutional Neural Network)	Global facility dataset (satellite images)	Classified 63 types of facilities (buildings, roads, airports, etc.) with ~83% accuracy, showing CNN's strength in image recognition.
3	AI-Driven Change Detection in Satellite Imagery: Enhancing GIS Applications	CNNs + GIS integration	Multi-source satellite data	AI detected urbanisation and deforestation more accurately than traditional techniques; results were integrated into GIS for real-time monitoring.
4	Continuous Urban Change Detection from Satellite Image Time Series	Transformers + Temporal Feature Refinement	Multi-year satellite time-series	Proposed continuous monitoring approach instead of before–after images; improved detection of gradual changes in urban areas.
5	Evidence-Based Inference and Quantification of Urban Expansion Using YOLOv8 and High-Resolution Satellite Imagery	YOLOv8 + Google Earth Pro + GIS	Greater Faridabad, India (2014–2025)	Detected buildings & roads with 98% accuracy; quantified ~450% urban growth over 11 years; provided GIS-ready outputs.
6	Deep Learning-Based Change Detection in Remote Sensing Images: A Review	CNN, RNN, GANs (survey)	Many datasets (SAR, multispectral, hyperspectral)	Summarized strengths & weaknesses of AI models; deep learning significantly outperforms older pixel-based techniques.
7	AI for Environmental Sustainability: Advances, Challenges, and Future Directions	Broad AI (ML, DL, GIS, IoT)	Global perspective	Highlighted AI use in climate modeling, biodiversity, pollution monitoring, renewable energy; discussed ethical and data challenges.
8	Urban Expansion Prediction and Land Use/Land Cover Modeling	Editorial overview, policy analysis, hybrid & AI techniques	Global overview / Special Issue	Highlights integration of AI in urban expansion research; emphasizes need for holistic sustainable frameworks and advances in hybrid modeling techniques

9	Monitoring and Forecasting of Urban Expansion Using Machine Learning-Based Techniques	SVM classification, Logistic Regression-Markov chain, LCM, Fuzzy TOPSIS	Gharbia Governorate, Egypt	Urban area projected to increase from 7.8% to 15.2% by 2018; prediction shows continued growth to 33% by 2048; high accuracy (OA > 93%, Kappa > 0.7)
10	Neural Network-Based Urban Change Detection with Deep-Temporal Multispectral and SAR Remote Sensing	Ensemble Recurrent CNN with multispectral & SAR data, synthetic labels	Rotterdam, Liège, Limassol (Europe), multiple datasets	Automated large-scale continuous urban change detection; improved accuracy and robustness through deep temporal modeling; efficient cross-mission modeling
11	Continuous Urban Change Detection from Satellite Images with Temporal Feature Refinement and Multi-Task Integration	ConvNet + Transformer + Markov network; multi-task learning	SpaceNet 7, WUSU, TSCD datasets (global, China)	Outperforms SOTA on continuous urban change detection; achieves higher F1, IoU, accuracy; preserves temporal dimension for improved change mapping
12	A Novel CNN-LSTM-based Approach to Predict Urban Expansion	CNN-LSTM (ConvLSTM), unsupervised semantic segmentation	Riyadh, Jeddah, Dammam (Saudi Arabia)	Achieves 92% classification accuracy; outperforms GAN methods on RMSE, MSE, SSIM, PSNR metrics; suitable for urban growth

Research Gaps

From the examined works, the following key gaps are revealed:

Complexity of Models – Models utilizing RNNs, Transformers, or both achieve extremely good rates of accuracy but require extremely large datasets and intensive computation that are too large for a mini-project.

GIS Integration Dependence – Other projects rely on integration with GIS systems to support real-time mapping and decision support. Although powerful, this adds implementation complexity not suitable for a lightweight system.

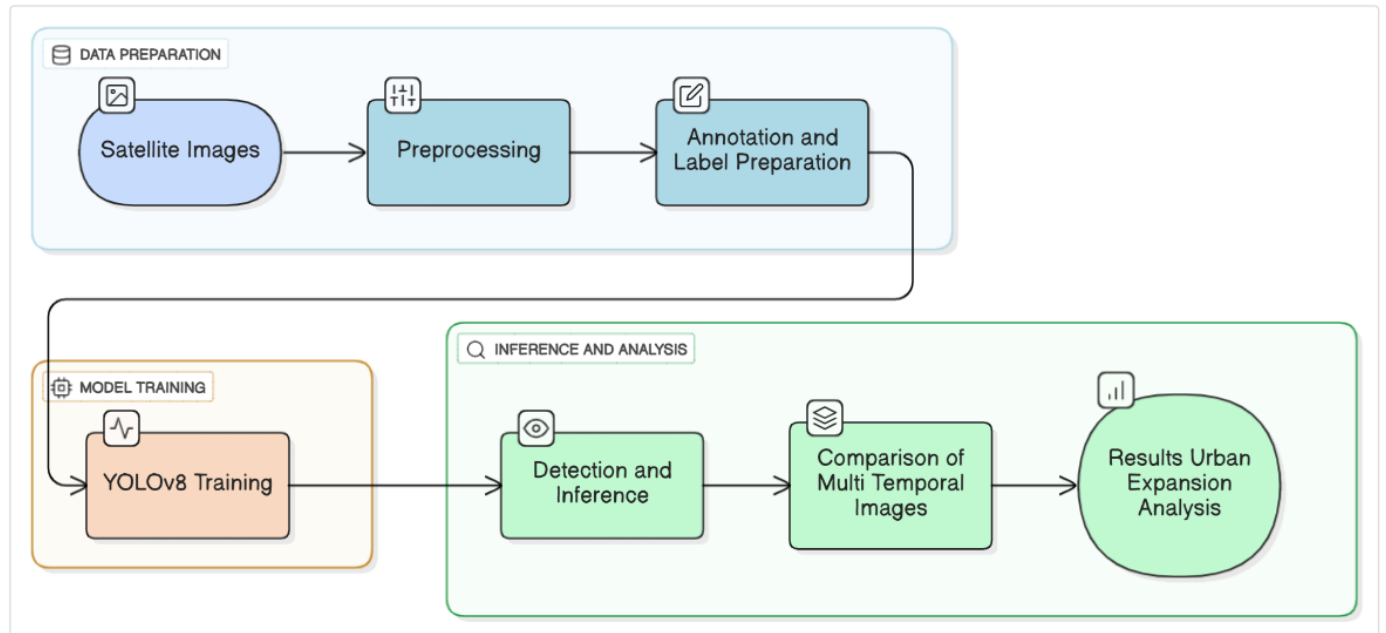
Temporal Data Handling – The camera systems function on multi-year long sequences of images and necessitate long-duration data handling and preprocessing pipelines.

Generalization vs. Practicality – Reviews mention that the deep learning models generalize well on different datasets, whereas simpler, direct-to-output approaches (e.g., object detection for a small dataset) are more translatable to demonstrable mini-projects.

Methodology

Satellite Images → Preprocessing → YOLOv8 Training → Detection → Change Detection → Results & Visualization

The methodology of the proposed system is divided into structured phases to ensure effective detection and analysis of urban expansion using satellite imagery. The overall workflow is shown in the block diagram below (you can draw/insert):



1. Data Collection

- Dataset used: SpaceNet 7 sample dataset or freely available satellite images from Google Earth.
- The dataset consists of multi-temporal images (images of the same area captured across different years).
- Focus feature: Buildings (as the primary indicator of urban expansion).

2. Preprocessing

- Convert images to a suitable format and resolution for YOLOv8 (e.g., .jpg or png with 640×640 pixels).
- Extract building footprint annotations and convert them into YOLO text-label format.
- Organize the dataset into train/validation/test splits.

3. Model Training

- Use YOLOv8n (Nano) or YOLOv8s (Small) model for lightweight training.
- Training steps:
 - Input: Preprocessed satellite images + YOLO-format labels.
 - Epochs: 50 (or based on hardware availability).
 - Output: A trained model (best.pt) that can detect buildings in unseen satellite images.

4. Object Detection & Inference

- Apply the trained YOLOv8 model on multi-temporal satellite images (e.g., 2017 and 2019 of the same city).
- The model outputs detected buildings with bounding boxes.
- Each detection includes confidence scores and coordinates.

5. Change Detection (Urban Expansion Analysis)

- Count the number of detected buildings in different years.
- Calculate growth rate or percentage increase in urban area:

$$\text{Urban Growth (\%)} = \frac{\text{Buildings in Year 2} - \text{Buildings in Year 1}}{\text{Buildings in Year 1}} \times 100$$

- Compare outputs visually (side-by-side images with bounding boxes) and quantitatively (growth statistics).

6. Results & Visualization

- Generate comparative images showing urban expansion across years.
- Provide numerical values like:
 - Number of buildings in Year 1 vs Year 2.
 - Percentage growth.
- Interpret the results in terms of urban sprawl, land-use change, and environmental impact.

Conclusion

The study rightly demonstrates the application of deep learning techniques through the YOLOv8 model on satellite scenes for monitoring urbanization. Through the recognition and assessment of the urban features such as buildings and built-up areas in multi-temporal scenes, the system provides evident indications of the patterns through which cities evolve over time.

The findings demonstrate the effectiveness of YOLOv8 in dealing with challenging satellite images and producing precise detections through minimal preprocessing. It is swifter than human inspection or conventional classification techniques and scalable and reliable when contrasted with them.

Even if such a mini-project involves only image processing and visualization of the result, the developed framework represents a good starting point for future extensions, e.g., the integration of GIS platforms, multi-class detection (roads, vegetation, water bodies), or real-time monitoring.

Finally, the research demonstrates that the analysis of satellite images using AI constitutes a useful and effective tool to monitor urbanization and help urban planners devise better urbanization plans for the future

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